

RailVolt: Distributed Traction System for Railway

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ABSTRACT

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Introduction: This paper presents *RailVolt*, a modular and scalable prototype aimed at retrofitting India's existing LHB railway coaches with a distributed traction system powered by microcontroller-based motorization. The motivation arises from limitations in the current single-locomotive traction system, which include high energy consumption, mechanical stress on couplers, and delayed braking responses.

Objectives: *RailVolt* is designed to motorize every third coach in a rake and synchronize them using UART communication, with the objective of reducing energy consumption by up to 30%, improving acceleration and braking smoothness, and minimizing wear on couplers. The system also aims to remain fully compatible with India's existing 25 kV AC overhead electrification and coach structures, avoiding the need for infrastructure overhauls.

Methods: The system utilizes an ESP32 controller as the master unit, which communicates with multiple Arduino Nano boards embedded in motorized coaches via UART. Each Nano regulates its respective motor using PWM signals, driven through L298N H-bridge motor drivers. Power is distributed through simulated pantograph-equipped feeder coaches and stabilized by onboard XL6009 buck-boost converters. Structurally, coaches are proportionally scaled using foam board and reinforced internally to support onboard electronics and motors.

Results: The prototype achieved real-time motor synchronization with sub-200 millisecond response times across coaches. Power delivery remained stable throughout the rake with negligible voltage drop. Visual displays confirmed accurate command execution, and test runs demonstrated smoother acceleration and braking profiles, validating the distributed traction approach.

Conclusions: *RailVolt* demonstrates the feasibility of retrofitting existing Indian rail coaches with synchronized distributed propulsion, providing an energy-efficient and cost-effective modernization pathway. The system integrates electrical control, mechanical design, and structural analysis into a unified working model that supports future extensions such as regenerative braking and AI-based control systems.

Keywords: Distributed propulsion, ESP32 control, LHB retrofit, motor synchronization, UART communication, Indian Railways

INTRODUCTION

Indian Railways operates one of the largest and most utilized railway networks in the world, transporting over 23 million passengers daily across a vast landscape and managing a fleet of more than 70,000 LHB (Linke Hofmann Busch) coaches. These coaches form the backbone of long-distance and intercity travel in India. Despite the scale of operations, most

of these trains rely on a single locomotive positioned at the front of the rake for propulsion. While this centralized traction system has been effective historically, it introduces several operational and mechanical inefficiencies when evaluated against the evolving demands of modern railway transport.

In a centralized configuration, the locomotive bears the entire responsibility of moving the full length of the train, which typically comprises 18 to 24 coaches. The tractive force is transmitted through the couplers that link each coach, resulting in high mechanical stress on the initial few couplers, especially during rapid acceleration and deceleration. This load concentration not only accelerates component wear but also contributes to sudden jerks and instability, affecting passenger comfort and increasing the likelihood of structural fatigue over time.

Moreover, the braking systems in conventional LHB rakes are predominantly pneumatic. These systems, while reliable, introduce delays in brake signal propagation from the locomotive to the rear coaches. This sequential actuation leads to inefficient and unbalanced braking, further amplifying longitudinal forces within the rake and increasing braking distance. In scenarios involving emergency stops or high-speed operation, these delays can pose serious safety risks.

While high-speed rail solutions such as bullet trains promise to address many of these performance challenges, their adoption in India remains limited due to substantial infrastructure requirements. Bullet trains demand dedicated tracks, specialized trainsets, high capacity signaling networks, and significant land acquisition. The estimated investment for such systems is exceedingly high, making them economically unviable for widespread implementation in the near future.

To address the shortcomings of conventional systems without overhauling the entire infrastructure, the RailVolt project introduces an innovative retrofit solution. Instead of replacing locomotives or building entirely new trainsets, RailVolt proposes motorizing alternate coaches, specifically, every third coach by integrating independent drive units and embedded microcontrollers. Each motorized coach can receive synchronized commands from a central controller (ESP32) using UART-based communication, enabling precise and coordinated traction.

The power required for propulsion is drawn from the existing 25 kV AC overhead electrification (OHE), simulated in the prototype through a 12V DC distribution line. Selected coaches function as power feeder units, stepping down and regulating the incoming voltage using buck-boost converters to supply controlled power to motorized units. This decentralized configuration mimics the behaviour of electric multiple units (EMUs), distributing propulsion across the train length and minimizing the strain on any single coupler or bogie.

RailVolt is designed to be scalable, modular, and fully compatible with existing Indian Railways infrastructure. It offers a cost-effective modernization pathway, improving energy efficiency, mechanical reliability, and passenger comfort without requiring significant investment in new corridors or rolling stock. By leveraging embedded systems,

distributed control logic, and modular power delivery, RailVolt provides a practical framework for retrofitting India's extensive fleet of LHB coaches, bringing them closer in performance to advanced rail systems at a fraction of the cost.

OBJECTIVES

The RailVolt project is driven by the need to modernize India's railway propulsion systems without overhauling existing infrastructure. Traditional single-locomotive traction systems introduce energy inefficiencies, mechanical stress on couplers, and passenger discomfort due to uneven force distribution. RailVolt addresses these limitations through a distributed propulsion architecture.

The core objectives of the project are:

- 1. Energy Optimization:**
To reduce overall energy consumption by an estimated 20 - 30% by distributing tractive load across multiple motorized coaches, thereby decreasing the burden on any single unit.
- 2. Enhanced Ride Quality:**
To minimize mechanical jerks and improve passenger comfort by enabling smoother acceleration and braking through synchronized, decentralized control.
- 3. Mechanical Stress Reduction:**
To reduce stress on couplers and structural components by applying propulsion and braking forces more evenly throughout the train.
- 4. Infrastructure Compatibility:**
To ensure full compatibility with existing Indian Railways infrastructure, particularly the 25 kV AC OHE system and LHB coach designs, avoiding the need for costly track or system modifications.
- 5. Scalability and Retrofit Potential:**
To develop a modular and scalable prototype that can be retrofitted into existing rakes with minimal structural changes, enabling practical large-scale adoption.

These objectives collectively aim to create a viable, low-cost, and energy-efficient alternative to centralized traction systems, providing a stepping stone toward smarter and more sustainable railway operations in India.

METHODS

The RailVolt system was developed as a functional prototype to simulate distributed traction in Indian Railways. The methodology integrates embedded control, motor actuation, power regulation, and mechanical construction. The design focuses on modularity, scalability, and real-world replication within the constraints of a scaled-down experimental setup.

1) Distributed Motor Control Architecture

At the heart of RailVolt's propulsion control is the ESP32 microcontroller, functioning as the central command unit. It generates real-time speed instructions which are transmitted via UART (Universal Asynchronous Receiver/Transmitter) to motorized coaches embedded with Arduino Nano boards. Each Arduino Nano decodes the command and regulates its assigned DC motor using PWM (Pulse Width Modulation) signals.

For motor driving, the system employs L298N dual H-bridge drivers, which allow bidirectional control—enabling both

forward and reverse motion. This configuration is suitable for simulating advanced features like coordinated braking or reverse traction behavior. While regenerative braking is not implemented in the current version, the hardware allows for its future integration.

2) Power Distribution and Voltage Regulation

To emulate the 25 kV AC Overhead Electrification (OHE) system used in Indian Railways, the prototype uses a 12V DC line routed through selected feeder coaches (C3 and C6) that mimic pantograph-equipped units. These coaches distribute power to the rest of the rake using externally routed connections.

Each motorized coach is equipped with an XL6009 buck-boost converter, which stabilizes the voltage to safe operating levels for both the microcontrollers and motors. The power architecture is bifurcated into two isolated circuits:

- One for logic control
- One for motor drive

This separation reduces electromagnetic interference and ensures signal integrity during UART communication.

3) Communication and Synchronization

Reliable inter-controller communication is critical for distributed traction. The ESP32's TX pin is daisy-chained to the RX pins of subsequent Arduino Nanos in each motorized coach. This configuration facilitates low-latency, one-directional command propagation.

The communication protocol includes:

- Delimiter bytes for packet boundaries
- Checksum validation for integrity
- Error-detection routines to handle transmission faults

This ensures synchronized motor actuation across the train, enabling uniform acceleration and braking without controller desynchronization.

4) Mechanical and Structural Design

Each coach is built using foam board, chosen for its favorable strength-to-weight ratio. The body houses onboard electronics within dedicated compartments and maintains balanced weight distribution through internal reinforcements made of balsa wood or light plastic frames. Motors are mounted within 3D-printed bogie assemblies, with axles aligned to simulate real coach mechanics.

Wiring is routed through pre-cut internal channels to preserve center-of-gravity alignment and avoid drag or imbalance during motion. Couplers between coaches are detachable, enabling modular testing of different configurations.

5) Sensor Integration and Validation Tools

To validate electrical and mechanical behavior, the prototype includes:

- ACS712 current sensors to measure motor power draw
- Accelerometers to record jerk and vibration during motion
- OLED and LCD displays to provide real-time UART signal monitoring

The system's CAD components, such as axle holders and motor mounts, were designed in SolidWorks and fabricated using Ultimaker Cura for 3D printing.

RESULTS

The RailVolt prototype was constructed and tested under controlled conditions to evaluate its performance in simulating distributed traction in railway systems. Both electrical and mechanical aspects were assessed to verify synchronization, power stability, communication reliability, and structural integrity.

1) Synchronized Motor Control Using UART

The system successfully maintained synchronized propulsion across motorized coaches (C1, C4, and C7) using UART-based serial communication. The ESP32 master controller transmitted commands to Arduino Nano units, which responded accurately without signal degradation or data loss. Motor activation across all units occurred within a consistent response time of under 200 milliseconds, confirming reliable real-time control.

2) Stable Power Distribution

The power delivery system, which simulates Indian Railways' 25 kV AC OHE using a 12V DC line, distributed voltage effectively across the rake. Pantograph-simulated feeder coaches (C3 and C6) supplied power to motorized coaches via external connections. Multimeter readings confirmed negligible voltage drop between adjacent coaches, and XL6009 buck-boost converters successfully stabilized the input voltage at each motor node.

3) Accurate Speed Feedback and Monitoring

OLED and LCD modules integrated into selected coaches displayed real-time speed and command data received via UART. Displayed values matched programmed inputs without observable lag, validating that the data parsing, communication, and display logic were functioning correctly.

4) Smooth Acceleration and Braking

During live test runs, the train demonstrated smoother transitions in acceleration and deceleration compared to conventional systems. Mechanical jerks, typically caused by centralized force application in single-locomotive setups, were significantly reduced due to distributed propulsion. This was confirmed both qualitatively through observation and quantitatively using accelerometer data.

5) Structural Performance

The modular foam-board coaches with embedded electronics retained mechanical stability throughout operations. No structural failures or displacement issues were observed, validating the mechanical design approach using lightweight materials reinforced internally for torque absorption and balance.

These results confirm that RailVolt achieves synchronized distributed traction with stable power flow and realistic mechanical behaviour, making it a viable platform for future-scale railway retrofitting and advanced control system experimentation.

DISCUSSION

The implementation of distributed traction in RailVolt demonstrates measurable advantages over traditional single-locomotive systems. By motorizing every third coach and synchronizing them via UART communication, the prototype successfully replicates the core dynamics of electric multiple units (EMUs) without requiring fundamental changes to India's existing railway infrastructure.

One of the most significant findings was the marked reduction in mechanical jerks during acceleration and braking, validating the concept of load distribution across multiple motorized units. This aligns with the theoretical benefits

of distributed propulsion, where tractive and braking forces are evenly applied along the rake, reducing stress concentrations at couplers and improving passenger comfort.

The control system's responsiveness—with motor activation delays consistently under 200 milliseconds—supports its viability for real-world applications. Furthermore, the UART-based communication protocol, augmented with packet-based integrity checks, proved reliable under continuous operation, highlighting the potential of low-cost microcontrollers like the ESP32 and Arduino Nano in embedded railway systems.

From a structural perspective, the use of lightweight foam-board coaches reinforced with balsa wood and integrated with 3D-printed bogies ensured that electronic components could be housed without compromising mechanical stability. This suggests that existing coaches could be retrofitted with minimal alteration, making RailVolt an attractive candidate for low-disruption modernization efforts.

When compared to high-speed rail or full EMU deployment, RailVolt offers a scalable, cost-efficient alternative that reuses existing assets. While the current prototype focuses on functional feasibility, future iterations could integrate features like regenerative braking, AI-based predictive control, and enhanced diagnostics for fault tolerance.

Overall, RailVolt bridges the gap between current infrastructure limitations and the need for smarter rail systems. It offers a pathway to incremental modernization that is both economically and technically feasible in the Indian railway context.

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